

2 (a) The moment of a force about a point is defined as the product of the force and the perpendicular distance of the line of action of the force from the point.

(b) Since the airplane is travelling at constant speed, it is in equilibrium

Vertically: $L = W = mg = (1.5 \times 10^5)(9.81) = 1.47 \times 10^6 \text{ N}$

Horizontally: $T = D = 8.0 \times 10^6 \text{ N}$

clockwise torque = anti-clockwise torque

$$T \times y = W \times (0.75)$$

$$(8.0 \times 10^5) y = (1.47 \times 10^6)(0.75)$$

$$y = 1.38 \text{ m}$$

- (c) As the airplane accelerates forward, the box will tend to move at its original velocity. Hence, the box will tend to move backwards relative to the airplane floor. Friction on the box by the floor therefore acts forward (to oppose this relative motion).

- (d) Applying principle of moments, taking pivot at the C.G.,

$$(2.0 \times 10^5) 10 + (2.0 \times 10^5) 20 = P (30 \cos 30^\circ)$$

$$P = 2.31 \times 10^5 \text{ N}$$